

16.4 The Cobweb Model

To illustrate the use of first-order difference equations in economic analysis, we shall cite two variants of the market model for a single commodity. The first variant, known as the *cobweb model*, differs from our earlier market models in that it treats Q_t as a function not of current price but of the price of the preceding time period.

the model Consider a situation in which the producer's output decision must be made one period in advance of the actual sale—such as in agricultural production, where planting must precede by an appreciable length of time the harvesting and sale of the output. Let us assume that the output decision in period t is based on the then-prevailing price P_t . Since this output will not be available for sale until period $(t + 1)$, however, P_t will determine not Q_{at} but $Q_{a,t+1}$. Thus we now have a "lagged" supply function¹

$$Q_{a,t+1} = S(P_t)$$

or, equivalently,

$$Q_{at} = S(P_{t-1})$$

When such a supply function interacts with a demand function of the form

$$Q_{dt} = D(P_t)$$

interesting dynamic price patterns will result.

Taking the linear versions of these (lagged) supply and (unlagged) demand functions, and assuming that in each time period the market price is always set at a level which clears the market, we shall have a market

¹ We are making the implicit assumption here that the entire output of a period will be placed on the market, with no part of it held in storage. Such an assumption is appropriate when the commodity in question is perishable or when no inventory is ever kept. A model with inventory will be considered in the next section.

model with the following three equations:

$$\begin{aligned} Q_{dt} &= Q_{dt} \\ (16.10) \quad Q_{dt} &= \alpha - \beta P_t \quad (\alpha, \beta > 0) \\ Q_{dt} &= \gamma + \delta P_{t-1} \quad (\gamma < 0; \delta > 0) \end{aligned}$$

By substituting the last two equations into the first, however, the model can be reduced to a single first-order difference equation as follows:

$$\beta P_t + \delta P_{t-1} = \alpha - \gamma$$

In order to solve this equation, it is desirable first to normalize it and shift the time subscripts ahead by one period [alter t to $(t+1)$, etc.]. The result,

$$(16.11) \quad P_{t+1} + \frac{\delta}{\beta} P_t = \frac{\alpha - \gamma}{\beta}$$

will then be a replica of (16.6), with the substitutions

$$y = P \quad a = \frac{\delta}{\beta} \quad \text{and} \quad c = \frac{\alpha - \gamma}{\beta}$$

Inasmuch as δ and β are both positive, it follows that $a \neq -1$. Consequently, we can apply formula (16.8'), to get the time path

$$(16.12) \quad P_t = \left(P_0 - \frac{\alpha - \gamma}{\beta + \delta} \right) \left(\frac{-\delta}{\beta} \right)^t + \frac{\alpha - \gamma}{\beta + \delta}$$

where P_0 represents the initial price.

the cobwebs Three points may be observed in regard to this time path. In the first place, the expression $(\alpha - \gamma)/(\beta + \delta)$, which constitutes the particular integral of the difference equation, can be taken as the equilibrium price of the model:¹

$$\bar{P} = \frac{\alpha - \gamma}{\beta + \delta}$$

Being a constant, this is a stationary equilibrium. Substituting $\bar{P}$ into our solution, we can express the time path P_t alternatively in the form

$$(16.12') \quad P_t = (P_0 - \bar{P}) \left(\frac{-\delta}{\beta} \right)^t + \bar{P}$$

This leads us to the second point, namely, the significance of the expression $(P_0 - \bar{P})$. Since this corresponds to the constant A in the Ab^t term, its

¹ The reader may verify this as follows: When price is in equilibrium, we must have $P_t = P_{t-1} = \bar{P}$. By setting $P_t = P_{t-1} = \bar{P}$ in (16.11) and solving for $\bar{P}$, we obtain $\bar{P} = (\alpha - \gamma)/(\beta + \delta)$. This procedure is, of course, the familiar one of finding a particular integral by the use of the simplest (constant) trial solution.

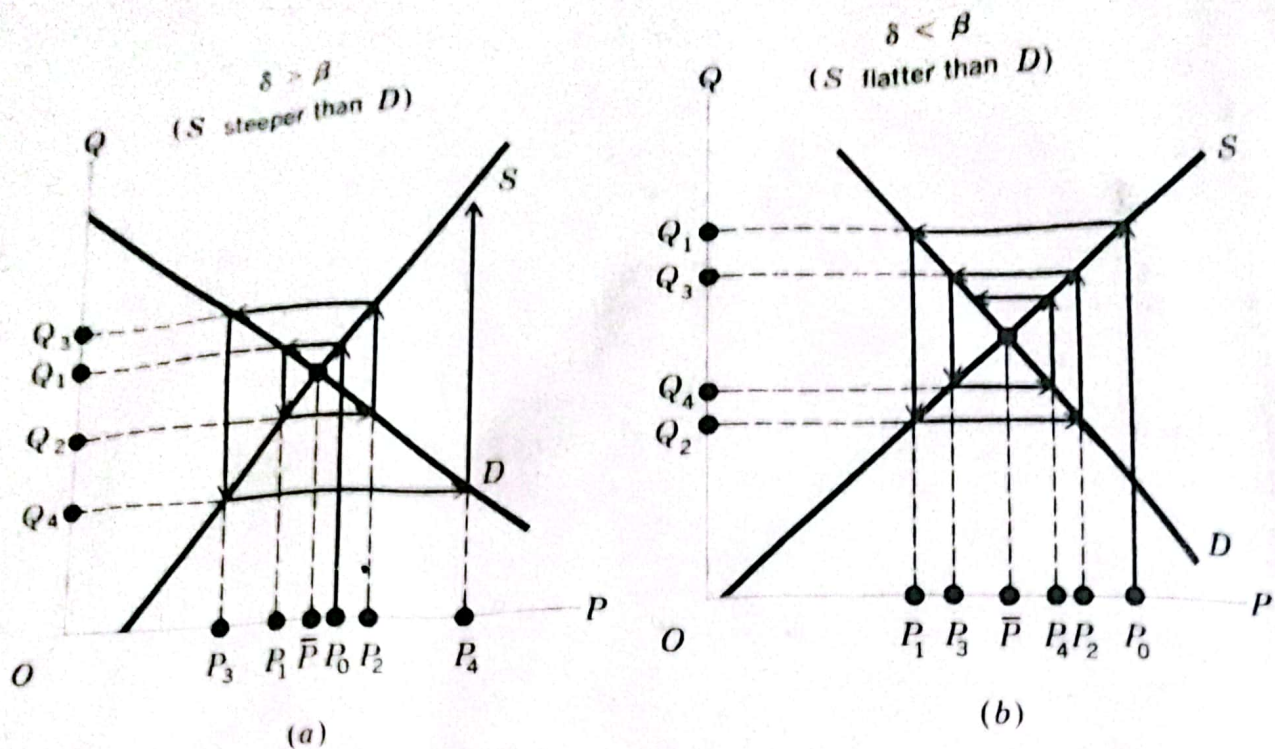


FIGURE 16.2

sign will bear on the question of whether the time path will commence above or below the equilibrium (mirror effect), whereas its magnitude will decide how far above or below (scale effect). Lastly, there is the expression $(-\delta/\beta)$, which corresponds to the b component of Ab' . Since our model specification has it that $\beta, \delta > 0$, we must have an oscillatory time path. It is this fact which gives rise to the cobweb phenomenon, as we shall presently see. There can, of course, arise *three* possible varieties of oscillation patterns in the model. According to Table 16.1 or Fig. 16.1, the oscillation will be

$$\left. \begin{array}{l} \text{explosive} \\ \text{regular} \\ \text{damped} \end{array} \right\} \text{ if } \delta \gtrless \beta$$

In order to visualize the cobwebs, let us depict the model (16.10) in Fig. 16.2. The second equation of (16.10) plots as a downward-sloping linear demand curve, with its slope numerically equal to β . Similarly, a linear supply curve with a slope equal to δ can be drawn from the third equation, if we let the Q axis represent in this instance a *lagged* quantity supplied. The case of $\delta > \beta$ (S steeper than D) and the case of $\delta < \beta$ (S flatter than D) are illustrated in diagrams a and b , respectively. In either case, however, the intersection of D and S will yield the equilibrium price $\bar{P}$.

When $\delta > \beta$, as in diagram a , the interaction of demand and supply will produce an explosive price path as follows. Given an initial price P_0

(here assumed above $\bar{P}$), we can follow the arrowhead and read off on the S curve that the quantity supplied in the next period (period 1) will be Q_1 . In order to clear the market, the quantity demanded in period 1 also must be Q_1 , which is possible if and only if price is set at the level of P_1 (see downward arrow). Now, via the S curve, the price P_1 will lead to Q_2 as the quantity supplied in period 2, and to clear the market in the latter period, price must be set at the level of P_2 according to the demand curve. Repeating this reasoning, we can trace out the prices and quantities in subsequent periods by simply following the arrowheads in the diagram, thereby weaving a "cobweb" around the demand and supply curves. By comparing the price levels, $P_0, P_1, P_2, \dots$, we observe in this case not only an oscillatory pattern of change but also a tendency for price to widen its deviation from $\bar{P}$ as time goes by. With the cobweb being woven from inside out, the time path is divergent and explosive.

By way of contrast, in the case of diagram b , where $\delta < \beta$, a similar weaving process will create a cobweb that is center-oriented. From P_0 , if we follow the arrowheads, we shall be led ever closer to the intersection of the demand and supply curves, where $\bar{P}$ is. While still oscillatory, this price path is convergent.

In Fig. 16.2 we have not shown a third possibility, namely, that of $\delta = \beta$. The procedure of graphical analysis involved, however, is perfectly analogous to the other two cases. It is therefore left to the reader as an exercise.

The above discussion has dealt only with the time path of P (that is, P_t); after P_t is found, however, it takes but a short step to get to the time path of Q . The second equation of (16.10) relates Q_{dt} to P_t , so that if (16.12) or (16.12') is substituted into the demand equation, the time path of Q_{dt} can be obtained immediately. Moreover, since Q_{dt} must be equal to Q_{st} in each time period (clearance of market), we can simply refer to the time path as Q_t rather than Q_{dt} . On the basis of Fig. 16.2, the rationale of this substitution is easily seen. Each point on the D curve relates a P_t to a Q_t pertaining to the same time period; therefore the demand function can serve to map the time path of price into the time path of quantity.

The reader will note that the graphical technique of Fig. 16.2 is applicable even when the D and S curves are nonlinear.

EXERCISE 16.4

- 1 On the basis of (16.10), find the time path of Q , and analyze the condition for its convergence.

- 2 Draw a diagram similar to those of Fig. 16.2 to show that, for the case of $\delta = \beta$, the price will oscillate regularly.
- 3 Given demand and supply for the cobweb model as follows, find the equilibrium price, and determine whether the equilibrium is stable:

(a) $Q_{dt} = 18 - 3P_t$ $Q_{st} = -3 + 4P_{t-1}$

(b) $Q_{dt} = 22 - 3P_t$ $Q_{st} = -2 + P_{t-1}$

(c) $Q_{dt} = 19 - 6P_t$ $Q_{st} = 6P_{t-1} - 5$